

CHAPTER 1 — STRING BASICS & DECLARATION

SLIDE 1 — WHAT IS A STRING?

MEET STRINGS!

You already know that a `char` can store **one character**.

But what if we want to store:

Hello

Ravi

Computer

 We need to store **multiple characters together**.

Simple Definition

A string is a collection of characters used to represent text.

LOOK AT THIS!

"Hello"

It contains:

H e l l o
↓ ↓ ↓ ↓ ↓
    

Each one is a **character**.

Together → **String** 

EASY ANALOGY

Think of a **word as a train** 

 H →  e →  l →  l →  o
↓
 "Hello"

 **One character** = One coach

 **Many characters together** = A string

★ REMEMBER!

Character → One 

String → Many   

SLIDE 2 — CHARACTER vs STRING

ARE THEY THE SAME?

Not exactly!

CHARACTER

A character represents **one** letter, number, or symbol.

'A'

👉 Uses **single quotes** ' ' .

Examples:

'B'

'7'

'@'

STRING

A string contains **multiple characters**.

"Hello"

👉 Uses **double quotes** " " .

Examples:

"Ravi"

"Hello World"

"12345"

⚡ EASY COMPARISON



Character



String

One character

Collection of characters

'A'

"Apple"

Single quotes

Double quotes

'7'

"12345"



MEMORY TRICK

'A' → ONE

"Apple" → MANY



DON'T MIX THEM!

'A' Character

"Apple" String

Think:

Single quotes → Single character

Double quotes → String



SLIDE 3 — CHARACTER ARRAY



WHERE DOES C STORE A STRING?

In C, a string is stored using a **character array**.



Simple Definition

A character array is an array used to store characters.

For example:

```
char name[] = "Hello";
```

C stores the characters separately:

Index → 0 1 2 3 4



THINK OF IT AS BOXES

    
H e | | o

Each box stores **one character**.

Together:

    
↓
"Hello"

WHY **char**?

Because a character array stores **characters**.

```
char name[];
```

 **char** → character type

 **name** → array name

[] → array

TAKEAWAY

String in C → Character Array 

SLIDE 4 — DECLARING & INITIALIZING A STRING

DECLARING A STRING

Declaration means **creating space to store the characters**.

```
char name[20];
```

BREAK IT DOWN

char →  Character type
name →  Array name
[20] →  Space for characters

Think:

```
char name[20];
```



We have created space for the string.

INITIALIZING A STRING

Initialization means **giving the string a value**.

```
char name[] = "Hello";
```

Now the boxes contain:



DECLARATION vs INITIALIZATION

```
char name[20];
```

 **Create space**

vs.

```
char name[] = "Hello";
```

 **Create + give a value**

MEMORY TRICK

 **Declaration** → "Give me space!"

 **Initialization** → "Put something inside!"

SLIDE 5 — STRING BASICS RECAP

YOU'VE MET STRINGS!

 REMEMBER THESE 5 THINGS!

1. STRING

A collection of characters used to represent text.

"Hello"

2. CHARACTER

One character.

'A'

3. CHARACTER ARRAY

Used to store characters together.

```
char name[];
```

4. DECLARATION

Creates space for the string.

```
char name[20];
```

5. INITIALIZATION

Gives the string a value.

```
char name[] = "Hello";
```

ONE BIG MEMORY TRICK!

 CHARACTER



One symbol

 STRING



Many characters

 CHARACTER ARRAY



Stores the characters

 DECLARATION



Creates space

 INITIALIZATION



Gives a value

 **SEE IT IN CODE**

```
char name[] = "Hello";
```

char →  Character type

name →  Name

[] →  Character array

"Hello" →  String value

 **LEVEL CLEARED!**

I can now identify, declare, and initialize a string in C! 🎉

Questions:

Q1. MCQ 

Which of the following is a string?

a) 'A'

b) "Hello"

- c) 25
- d) 3.14

💡 Hint: A string is a collection of characters written using double quotes.

📖 Glossary – string → A collection of characters used to represent text. | characters → Individual letters, numbers, or symbols.

Answer: ☒ B) "Hello"

Q2. MCQ

What is a character array used for in C?

- a) To store decimal numbers
- b) To store characters together
- c) To display output
- d) To perform calculations

💡 Hint: Think about where a string is stored in C.

📖 Glossary – character array → An array used to store characters. | store → Keep information in memory.

Answer: ☒ B) To store characters together

Q3. Choose the Correct Code

Which code correctly declares a string named name?

a)

```
int name[20];
```

b)

```
float name[20];
```

c)

```
char name[20];
```

d)

```
string name[20];
```

💡 Hint: A string is stored using a character array.

📖 Glossary – declares → Creates a variable or array for use in a program. | character array → An array used to store characters.

Answer: ☒ C) char name[20];

Q4. Choose the Correct Code 🖥️

Which code correctly initializes a string with "Hello"?

a)

```
char word = "Hello";
```

b)

```
char word[] = "Hello";
```

c)

```
int word[] = "Hello";
```

d)

```
char word[] = 'Hello';
```

💡 Hint: A string uses double quotes and is stored in a character array.

📖 Glossary – initializes → Gives a variable or array an initial value. | double quotes → " " used to represent a string.

Answer: ☒ B) char word[] = "Hello";

Q5. Fill in the Missing Code

Complete the code to create a character array that stores "Ravi".

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    -----
```

```
    return 0;
```

```
}
```

- a) `int name = "Ravi";`
- b) `char name = "Ravi";`
- c) `char name[] = "Ravi";`
- d) `float name[] = "Ravi";`

 Hint: Use `char` because the string is made up of characters.

 Glossary – character array → An array used to store characters. |
string → A collection of characters used to represent text.

Answer:  C) `char name[] = "Ravi";`